
Quiz: Model Fine-Tuning & LLM Optimization

Section 1: Conceptual MCQs

1. **What is model fine-tuning?**
 - A. Training a model from scratch
 - B. Adjusting a pre-trained model on new data
 - C. Only changing prompts
 - D. Increasing model size

Answer: B → Adjusting a pre-trained model

2. **Which scenario is best suited for prompting instead of fine-tuning?**
 - A. Domain-specific chatbot
 - B. Repetitive structured output task
 - C. One-time query with general knowledge
 - D. Training on private dataset

Answer: C → One-time general query → prompting best

3. **What is the main advantage of fine-tuning over prompting?**
 - A. Faster inference always
 - B. Better control over task-specific outputs
 - C. No need for data
 - D. Lower memory usage

Answer: B → Better task-specific control

4. **What does LoRA stand for?**
 - A. Low Rank Adaptation
 - B. Linear Representation Adjustment

- C. Layer Optimization and Reuse Algorithm
- D. Learning Rate Adjustment

Answer: A → Low Rank Adaptation

5. In LoRA, what is actually trained?

- A. Entire model weights
- B. Only input tokens
- C. Small low-rank matrices
- D. Output layer only

Answer: C → Small low-rank matrices

6. Why is LoRA considered parameter-efficient?

- A. It reduces dataset size
- B. It updates only a small subset of parameters
- C. It removes layers
- D. It compresses input

Answer: B → Only small subset updated

7. What is the role of a tokenizer in fine-tuning?

- A. Train model weights
- B. Convert text into tokens
- C. Evaluate model
- D. Store embeddings

Answer: B → Text → tokens

☐

8. Which step comes first in a fine-tuning workflow?

- A. Model evaluation
- B. Data collection & cleaning
- C. Training
- D. Deployment

Answer: B → Data collection first

9. What is overfitting in fine-tuning?

- A. Model performs well on all data
- B. Model memorizes training data but fails on new data

- C. Model ignores training data
- D. Model becomes faster

Answer: B → Memorizes training, fails on new data

10. Which metric is commonly used for classification tasks?

- A. BLEU score
- B. Accuracy
- C. Perplexity
- D. Latency

Answer: B → Accuracy

Section 2: True / False

11. Fine-tuning always requires large datasets.

False

12. Prompting does not change model weights.

True

13. LoRA modifies all layers of the model.

False

14. Validation data is used to tune hyperparameters.

True

15. Fine-tuned models can outperform base models on niche tasks.

True

Section 3: Scenario-Based Questions

16. Scenario:

You need a chatbot specialized in legal documents with consistent tone and formatting.

 What should you choose?

- A. Prompting
- B. Fine-tuning
- C. Zero-shot learning
- D. Random initialization

Answer: B → Fine-tuning (legal domain + consistency)

17. Scenario:

You want quick results for generating marketing slogans without training.

👉 Best approach?

- A. Fine-tuning
- B. Prompt engineering
- C. LoRA
- D. Training from scratch

Answer: **B** → Prompt engineering (fast, no training)

18. Scenario:

You have limited GPU resources but want to adapt a large model.

👉 Best method?

- A. Full fine-tuning
- B. LoRA
- C. Increase batch size
- D. Add more layers

Answer: **B** → LoRA (low GPU)

19. Scenario:

Your model performs well on training data but poorly on test data.

👉 What is the issue?

- A. Underfitting
- B. Overfitting
- C. Data leakage
- D. Tokenization error

Answer: **B** → Overfitting

20. Scenario:

You are comparing two models for summarization.

👉 Which metric is most relevant?

- A. Accuracy
- B. BLEU / ROUGE
- C. Latency only
- D. Token count

Answer: **B** → BLEU / ROUGE

Section 4: Short Answer

21. Explain one difference between prompting and fine-tuning.

Answer : ☐ **Prompting** → model change nahi hota

☐ **Fine-tuning** → model weights update hote hain

22. What is one benefit of using LoRA?

Answer : ☐ **Less computation + faster training**

23. List any two steps in a fine-tuning workflow.

Answer : ☐ ☐ **Data collection**

☐ **Training**

24. What is the purpose of validation data?

Answer : ☐ **Model tuning & overfitting check**

25. Name one metric used to evaluate LLMs.

Answer : ☐ **BLEU / ROUGE / Perplexity**

Section 5: Practical Thinking

26. You are fine-tuning a model and loss is not decreasing. What could be one reason?

Answer : ☐ ☐ **Wrong Learning rate**

☐ **data issue**

27. Why might fine-tuning be expensive?

Answer : ☐ **GPU + large model training**

28. When should you avoid fine-tuning?

Answer : ☐ ☐ **Task should be simple**

☐ **Prompting**

29. What happens if tokenizer is mismatched with the model?

Answer : ☐ **Wrong inputs → model output garbage**

30. How does LoRA reduce computational cost?

Answer : ☐ **Only small matrices are trained not full models.**